FABRICFLOW RESEARCH PROJECT

Component 3
Complete Technical Explanation

Emergency Detection, Production Risk Prediction and Recovery Management

Component 3 දෛනික නිෂ්පාදන දත්ත monitor කර current risk, overall order risk සහ ඉදිරි දින තුනේ early warnings හඳුනාගෙන, explainable recovery plan එකක් ලබාදෙන hybrid intelligent component එකකි.

Project: FabricFlow Garment Production Management System
Component: C3 — Emergency Detection & Production Risk Management
Document date: 21 October 2026

Contents

1. Component 3 overview
2. End-to-end workflow
3. C2 to C3 integration
4. Daily input data
5. Analysis pipeline
6. Feature engineering
7. Active ML models
8. Early-warning mechanism
9. Hybrid rules and severity
10. Recovery planning
11. Outputs
12. Saving and next working day
13. Database design
14. Monitoring and verification
15. Recovery history
16. C3 to C4 integration
17. Training and evaluation
18. Security and plant isolation
19. Limitations
20. Panel questions and answers

1. Component 3 overview

> **Component 3 – Emergency Detection and Production Risk Management**
> Confirmed bulk order එකක් දෛනිකව monitor කර production emergency සහ schedule risk කලින් හඳුනාගෙන, managerට ක්‍රියාත්මක කළ හැකි recovery recommendation ලබාදීම මෙහි අරමුණයි.

Component 3 මගින් සිදු කරන ප්‍රධාන කාර්යයන්:

- අද දවසේ production risk type එක හඳුනාගැනීම.
- සම්පූර්ණ order එක deadline risk එකකදැයි තක්සේරු කිරීම.
- ඉදිරි production days තුනේ machine, quality සහ schedule emergencies predict කිරීම.
- Risk severity සහ alert routing තීරණය කිරීම.
- Workers, machines සහ overtime සීමාවන්ට ගැළපෙන recovery plan එකක් සැදීම.
- Daily monitoring, verified outcomes සහ recovery history save කිරීම.
- Completed order එකේ aggregated risk data Component 4 වෙත ලබාදීම.

2. End-to-end workflow

C2 bulk order confirmation

↓

Admin assigns the order to a plant

↓

Plant Manager selects an assigned order

↓

C2/order details auto-fill in C3

↓

Daily live production data is entered

↓

Five ML artifacts + deterministic rules run

↓

Risk, early warnings and recovery plan are produced

↓

Official daily record and recovery history are saved

↓

Completed-order data is aggregated and passed to C4

3. C2 to C3 integration

C2 bulk-production plan එක `bulk_orders.c2_result_json` තුළ save වෙනවා. Admin order එක factory එකකට assign කළ විට plant, allocated quantity, daily commitment සහ capacity snapshot `order_plant_allocations` table එකේ save වෙනවා.

Plant Manager login වූ පසු `GET /plant/orders` endpoint එක JWT token එකේ plant ID අනුව orders filter කරනවා. ඒ නිසා එක් manager කෙනෙකුට තමන්ගේ plant එකට assigned orders පමණක් පෙනෙනවා.

Order එක select කළ විට auto-fill වන fields:

- Bulk Order ID, Style ID සහ Buyer name
- Assigned plant සහ plant location
- Allocated quantity සහ capacity-based daily commitment
- Approved date සහ buyer-required date
- Total production, cutting සහ sewing days

4. Plant Manager enters daily live data

Input	Purpose
Actual daily output	අද නිෂ්පාදනය කළ actual quantity එක.
Cumulative completed quantity	Order එකෙන් මේ දක්වා complete කළ මුළු quantity එක.
Daily damage quantity	අද damage/quality loss quantity එක.
Maximum allowed damage	Accept කළ හැකි daily quality limit එක.
Machine breakdown count	අද unavailable/failed machines ගණන.
Worker-shortage count	Plan එකට සාපේක්ෂව අඩු workers ගණන.
Recovery capacity	Add කළ හැකි workers, backup machines සහ overtime limits.

5. Analyse Risk & Build Recovery Plan

Frontend එක `POST /api/component3/predict` call කරනවා. Backend pipeline එක:

```
Input validation
  → Feature engineering
  → Current-risk prediction
  → Overall-order-risk prediction
  → Next-three-day early warnings
  → Deadline and severity rules
  → Capacity-aware recovery calculation
  → Final structured JSON response
```

> **Important:** Analyse button එක preview result එක ලබාදෙනවා. Official database record එක සෑදෙන්නේ **Save Daily Record** කළ පසු පමණයි.

6. Validation and feature engineering

Model call කිරීමට පෙර dates, quantities, ranges සහ cumulative totals validate කරනවා. Raw inputs වලින් model-ready features හදනවා:

- Output gap සහ gap percentage
- Damage ratio සහ output-to-commitment ratio
- Remaining quantity සහ remaining working days
- Day progress සහ required daily rate
- Cutting/sewing phase flags
- Deadline calendar days සහ available working days
- Commitment slack සහ slack percentage

```
Daily commitment = 500
Actual output    = 400
Output gap       = 500 − 400 = 100
Gap percentage   = 100 / 500 × 100 = 20%
```

Training සහ runtime inference දෙකම එකම shared feature-building code භාවිත කරන නිසා training-serving mismatch අවම වෙනවා.

7. Active machine-learning models

Artifact	Algorithm	Responsibility
`c3_model1_risk_type_v2.pkl`	Random Forest	Current production risk type සහ confidence.
`c3_model2_order_risk_v3.pkl`	Random Forest	Deadline-aware overall order-risk probability.
`c3_early_warning_machine_breakdown_v2.joblib`	Random Forest	ඉදිරි දින තුනේ machine-breakdown warning.
`c3_early_warning_quality_limit_v2.joblib`	Logistic Regression	ඉදිරි දින තුනේ quality-limit warning.
`c3_early_warning_output_schedule_risk_v2.joblib`	Logistic Regression	ඉදිරි දින තුනේ output/schedule warning.

Runtime request එකකදී models retrain වෙන්නේ නැහැ. කලින් train කළ artifacts `joblib.load()` මගින් memory එකට load කර `predict()` සහ `predict_proba()` භාවිත කරනවා.

Current-risk categories

No Issue | Machine Breakdown | Worker Shortage | Quality Issue | Commitment Too Low | Minor Delay | Production Failure

8. Early-warning mechanism

Early-warning inference එක current day එක සමඟ previous daily monitoring history එකත් භාවිත කරනවා. එමඟින් ඉදිරි production days තුන තුළ subtype emergency ඇතිවීමේ probability එක ලබාදෙනවා.

Logistic Regression model එක feature weights එකතු කර sigmoid function එකෙන් probability එකක් ලබාදෙනවා:

```
z = intercept + feature₁×weight₁ + feature₂×weight₂ + ...
Probability = 1 / (1 + e⁻ᶻ)
```

Universal 50% threshold එකක් assume කරන්නේ නැහැ. Order-grouped validation මත Macro-F1 උපරිම වන target-specific calibrated thresholds භාවිත කරනවා.

9. Hybrid ML and deterministic rules

C3 pure ML system එකක් නොව hybrid system එකක්:

ML responsibilities	Rule-based responsibilities
Risk type සහ confidence	Output-gap severity
Overall order-risk probability	Damage-limit validation
Future-warning probabilities	Deadline exposure සහ completion checks
Historical pattern recognition	Alert routing සහ safe recovery constraints

Severity levels: **No Risk, Minor, Moderate, High, Critical**. Output gap, incident type, damage, remaining quantity සහ deadline margin එකට එකතු කර severity එක හදනවා.

10. Recovery planning

Recovery plan වෙනම ML model එකක් නොව explainable deterministic capacity engine එකකි. එය remaining quantity, available days, required rate, workers, machines සහ overtime limit සලකා බලනවා.

Possible recommendations:

- Continue the current production plan.
- Increase working hours within the declared overtime limit.
- Reassign available workers.
- Activate feasible backup machines.

- Improve daily production efficiency.
- Escalate or reallocate when the deadline cannot be recovered locally.

මෙය rule-based කිරීමෙන් recommendation එක explainable, capacity-safe සහ physically executable වෙනවා.

11. Component 3 output

Final structured output එකේ sections:

- Order summary සහ daily production position
- Current risk type, confidence සහ severity
- Overall order-risk level සහ probability
- Next-three-day early warnings
- Schedule forecast සහ order progress
- Alert recipients සහ recommended action
- Capacity-aware recovery plan සහ assumptions

```
Current risk: Worker Shortage
Severity: Moderate
Overall order risk: High
Order-risk probability: 74%
Schedule status: Behind target
Recommendation: Add workers and use controlled overtime
```

12. Saving an official daily record

`POST /plant/orders/<order_id>/log` endpoint එක order assignment verify කර, next working day calculate කර, C2 data සහ live inputs combine කර C3 analysis officially run කරනවා.

```
Analyse Risk = preview only
Save Daily Record = analysis + official database persistence
```

Next working day calculation:

```
Next working day = MAX(saved working_day_no) + 1
```

Future production date එකක් save කිරීමට system එක ඉඩ දෙන්නේ නැහැ.

13. Database design

Main operational database — `garment.db`

`daily_logs` table එකේ order ID, plant ID, production date, daily inputs සහ full C3 result එක `c3_result_json` ලෙස save වෙනවා. මෙම copy එක integrated workflow සහ C4 aggregation සඳහා භාවිත

කරනවා.

C3 audit database — `instance/component3_tracking.db`

Canonical monitoring records, prediction input/output, supervisor verification, incidents, decisions සහ recovery outcomes මෙහි save වෙනවා.

Databases දෙක `bulk_order_id`, plant key සහ production date මඟින් සම්බන්ධ වෙනවා. Form draft එක browser local storage එකේ තිබිය හැකි නමුත් එය official record එකක් නොවේ.

14. Monitoring history and supervisor ground truth

Monitoring History page එකෙන් daily records, predicted risk, early warning, verification status සහ recorded user බලන්න පුළුවන්. Model prediction එක ground truth ලෙස භාවිත කරන්නේ නැහැ.

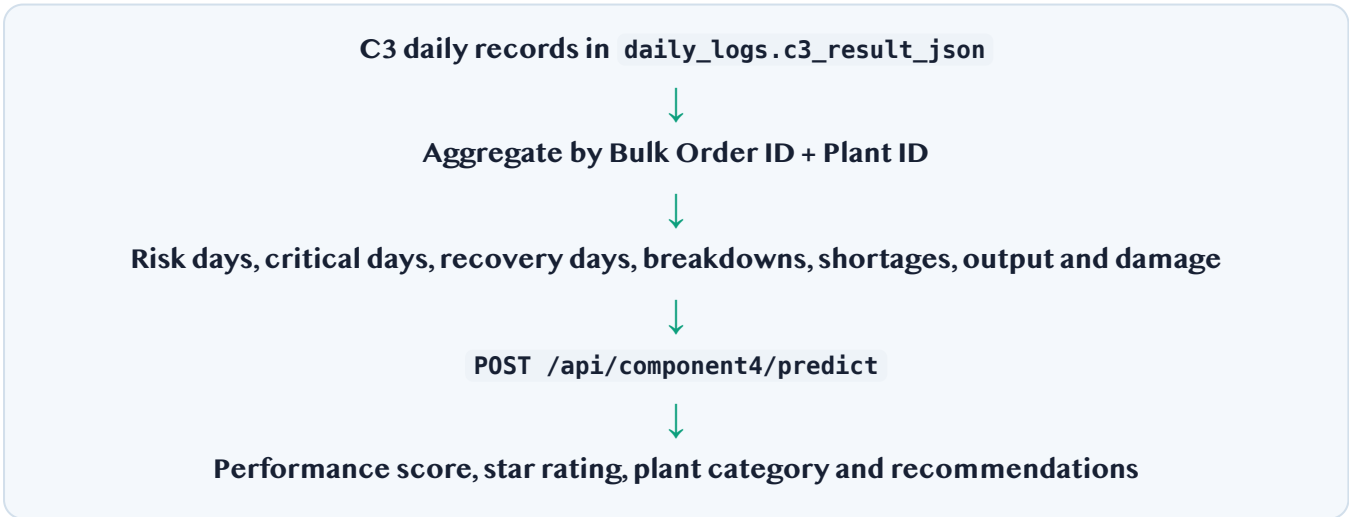
```
Model prediction ≠ Ground truth
Supervisor-verified actual outcome = Ground truth
```

Supervisor actual emergency occurred ද, actual type එක, verified-by user සහ evidence notes save කරනවා. මෙය future retraining එකේ self-label feedback loop එක වැළැක්වීමට වැදගත්.

15. Recovery History

Recovery History එකේ order, plant, incident type, severity, selected decision, responsible person, incident status, recovery outcome සහ notes track කරනවා. එමඟින් recommendation එක ක්‍රියාත්මක කළාද සහ successful වුණාද audit කරන්න පුළුවන්.

16. C3 to C4 integration



C4ට ලැබෙන ප්‍රධාන C3 feature එක `risk_count_from_component3` ය. C3 raw daily prediction එකක් direct යවන්නේ නැහැ; completed order එකේ සියලු daily records aggregate කරනවා.

17. Training, splitting and evaluation

Models offline training scripts වලින් train කරනවා. Daily API request එකකදී retraining සිදු නොවේ.

Evaluation element	Implementation
Grouping boundary	`Bulk_Order_ID`
Outer evaluation	Leave-One-Group-Out
Inner tuning	GroupKFold + GridSearchCV
Primary selection metric	Macro-F1
Reason	Emergency targets are imbalanced; Accuracy alone can hide minority-class failure.

එකම order එකේ daily rows train සහ test දෙකටම නොවැටෙන නිසා order-level data leakage වැළකෙනවා.

Active output	Model	Accuracy	Macro-F1
Current risk type	Random Forest	81.77%	90.08%
Overall order risk V3	Random Forest	99.83%	99.25%
Machine warning	Random Forest	86.21%	84.05%
Quality warning	Logistic Regression	66.67%	51.18%
Output/schedule warning	Logistic Regression	79.31%	63.75%

18. Security and plant-wise isolation

- JWT token එකෙන් user ID, role සහ plant ID ගන්නවා.
- Plant Managerට තමන්ගේ plant එකට assigned orders පමණක් ලැබෙනවා.
- Order assignment database එකෙන් නැවත verify කර daily log save කරනවා.
- Monitoring records සහ incidents plant scope එකට filter කරනවා.
- Buyerට internal C3 operational records access නැහැ.

19. Limitations

- Prediction quality එක daily input accuracy මත රඳා පවතී.
- Manual data entry මගින් errors සහ delays ඇතිවිය හැකියි.
- Rare emergencies සඳහා independent real-world examples සීමිතයි.
- Early-warning models තවත් unseen plants/orders මත validate කළ යුතුයි.
- Worker/machine availability update නොවුණොත් recovery plan practical නොවිය හැකියි.
- මෙය decision-support system එකක්; human managerගේ final decision එක replace නොකරයි.

20. Likely panel questions and concise answers

Q: What is Component 3?
A: Component 3 is the daily emergency-detection and production-risk-management component. It predicts current and future risks and generates an explainable recovery plan.

Q: How does C2 connect to C3?
A: C2 planning results and plant allocations are stored in the shared database. The authenticated plant manager selects an assigned order, and its dates, quantity, commitment and production-day details automatically populate C3.

Q: How many active ML artifacts are used?
A: Five: two main risk models and three next-three-day early-warning models.

Q: Why use Random Forest?
A: It captured nonlinear interactions among output, damage, resources and schedule features and achieved the strongest grouped Macro-F1 for current and overall risk.

Q: Why use Logistic Regression?
A: The regularised linear model generalised better for selected small and imbalanced early-warning targets and provides stable probabilities.

Q: Why is Macro-F1 important?
A: Accuracy can appear high when one class dominates. Macro-F1 gives emergency and non-emergency classes equal importance.

Q: How was leakage prevented?
A: Complete bulk orders were kept together using Bulk Order ID as the grouping boundary. An outer test order could not influence training, tuning or threshold selection.

Q: Is the recovery plan generated by ML?
A: No. ML detects risk; the recovery engine uses deterministic, capacity-aware calculations so the recommendation remains explainable and executable.

Q: How does C3 connect to C4?
A: Saved C3 daily records are aggregated by order and plant after completion. Risk counts and operational KPIs are then sent to C4 for performance scoring.

Q: Does saving a daily record retrain the model?
A: No. Runtime saves operational evidence. Retraining is a separate controlled offline process using verified outcomes.

Q: Why are there two databases?
A: The main database supports the integrated operational workflow, while the C3 tracking database preserves research-grade monitoring, verification and recovery audit history.

One-minute presentation answer

> “My Component 3 is a hybrid production-risk monitoring and emergency-management module. Component 2 assigns a confirmed bulk order to a plant and provides its production plan, dates and daily commitment. The plant manager records actual daily output, damage, breakdowns and worker shortages. Five active ML artifacts predict the current risk type, overall deadline risk and three categories of early warnings. These predictions are combined with deterministic deadline and capacity rules to determine severity and generate an explainable recovery plan. Official daily records, verified outcomes and recovery actions are stored by order and plant. After completion, aggregated Component 3 risk data is passed to Component 4 for plant-performance analysis.”

Generated from the integrated FabricFlow Component 3 implementation and project documentation.